

DeepTutor



“ ” “ ”

aicourse.top / Agentic Engineering

2026-08-23 · DeepTutor v1.5.16 · commit 8515dfd...

NOT_ASSESSABLE ·

DeepTutor

Tutor

DeepTutor

AI Tutor RAG Memory

P0/P1/P2



1. “AI Tutor” Agentic Engineering
RAG Prompt Software Engineering
[L5][L7][L8]
2. AI Tutor “ ”
“ ”
3. RAG RAG BM25
/ / GraphRAG PageIndex D
DeepTutor [D4][D5][D6]
4. DeepTutor “ ”
LLM judge [P1][P2]
5. “ ” Tutor
“ ” [L2][L3]
- “ Tutor DeepTutor ” RAG

1.

[README](#) [manifest](#) [DeepTutor v1.5.16](#) [README](#) [AGENTS](#).
[md](#) [RAG](#) [Memory](#) [Mastery](#) [Question](#) [Book](#) [Sandbox](#) [DeepTutor v3](#) [TutorBench](#)
[arXiv](#) [work-in-progress](#) [technical report](#) [\[D2\]\[P1\]](#)

- README
-
- aicourse.top DeepTutor

			NOT_ASSESSABLE	“available”	“ ”
[L1][L7]	DeepTutor	LLM judge	45	10	
	[P2]				

2.

2.1

□□	□□□□□	□□□□□□
	NOT_ASSESSABLE	“ ” “ ”
	TOP_LEVEL_COURSES available	release gate
/	Cursor soon Claude Grok	“ ” “Tutor ”
	S00-S10	

DeepTutor A AI Agentic Engineering Codex
GitHub Prompts Software Engineering RAG MCP Make Money with Codex Claude Income
available README Available course “ ” [L1][L7]

- Next.js [L1][L2]
- localStorage [L3]
- RAG fail-closed [L5]
- Agentic Engineering harness RAG capstone [L1][L5]

2.2 AI Tutor

DeepSeek
Tutor [L3][L4]
RAG Retrieval Lab “top-k”
“ RAG” RAG “ ” [L5][L6]

2.3

S00-S10 V3.1 Python LLM RAG AI Tutor
[L8]
Tutor policy Cursor Lesson 13 “ — — ”
Cursor soon Tutor [L7][L9]

2.4

□□	□□□□□□□□	DeepTutor v1.5.16	□□□□	□□□□□
	manifest QA	Book proposal spine typed blocks drift	Book	AI
Tutor	Tutor Cursor policy	/ pipeline	investigate-solve-write	
RAG	RAG	provider LlamaIndex BM25 vector RRF	SKG graph dense	
	localStorage learner model	L1/L2/L3 Memory Mastery	DPM Trace Forest schema	observation
	release checks Tutor outcome		LLM judge 45	→ →
/		provider		kill switch

DeepTutor A

3. DeepTutor

3.1 “ ”

DeepTutor Chat Quiz Research Visualize Solve Mastery Path Immersive R
eading “ loop” Question Research Visualize Math Animator Book
pipeline BookEngine BaseCapability [D1][D3]

3.2 RAG

RAG factory provider LlamaIndex PageIndex Cloud PageIndex OSS GraphRAG LightRAG L
ightRAG Server IMA provider MarginNote Obsidian capability/
factory provider LlamaIndex BM25 reciprocal-rank fusion
chunk id provider [D4][D5][D6]
needs_reindex version-N LlamaIndex embedding signature
provider embedding [D4][D6]

3.3 Memory

DeepTutor Memory L1 JSONL L2 Markdown L3
profile recent scope preferences Markdown L2 entity/trace ID

L3	/L2	L3 claim	L2 entry id	README	“	”
[D1][D7]						
DPM	Trace Forest		/ /	dense embedding	memory agents	
L1/L2/L3	schema			/		
Memory	[P2][D7]					

3.4

DeepTutor	Mastery Path					
BKT	IRT		[D12]			
Explore	Plan Quiz		/schema	repair	best-effort	
issues	validator	/	pipeline		Evidence	
Correctness	Pedagogy	schema	[D8][P2]			

3.5 Living Book

Book Engine		proposal	spine	typed blocks	
		partial/error	schema	source_anchors	
RAG	[D1][D9]				
“	”	fail-closed	source-anchor	—	

4. DeepTutor

DeepTutor	Static Knowledge Grounding SKG				Dynamic Personal Memory DPM			
	Trace Forest				memory agents			
					v1.5.16			
					[P1][P2]			
TutorBench	30	5	90	270	10	1-5	2	DeepTutor O
Q	3.91	Naive Tutor	3.53	+10.76%	Self-Refine	3.57	OQ	DeepTutor baseline
	SKG		groundedness/		/	DPM		[P2]
	DeepTutor							
	Tutor							
	/LLM judge				Book Partners			
	personalization		SKG	DPM	TutorBench			
harness	transcripts	judge records		[P1][P2]				

5. “AI Tutor” ”

RAG Agentic Engineering S04/S05/S06/S08 Cursor Lesson 13
Tutor API Tutor [L8][L9]

□□	□□□□□	□□□□□□	□□□□
1. Tutor		Agentic Engineering /	Tutor
2.		manifest RAG sources/checks	course-learning-contract.json
3. RAG		RAG 12 Retrieval Lab	
4.		Prompt Agentic loop	Tutor policy
5. Memory		context/memory	/
6.		quiz/capstone eval	Evidence/Correctness/Pedagogy gates
7. Tutor		Agent eval RAG eval	/
8.		Security Responsible AI	

DeepTutor A
capstone 50–100 retrieval gold queries 20–3
0 tutor dialogue cases protected outcome items / /

6. AI Tutor

6.1

- **Explain**
- **Coach** → → → →
- **Practice** “ ”

6.2

fail-closed

Cursor

[\[L5\]\[L9\]](#)

6.3

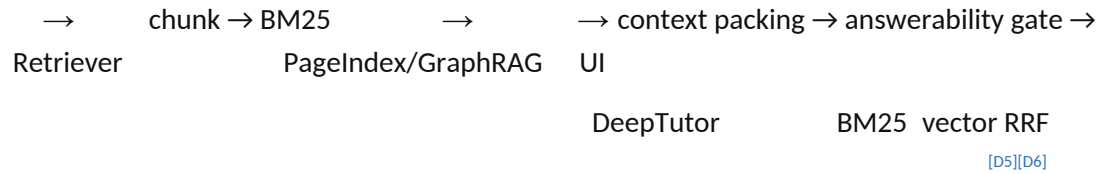
” “ `concept_id` ” “ ” “ ” / “

6.4

Tutor “ ” “ ” “ ” /

7. RAG

7.1



7.2

Tutor-enabled

- `course_id` `locale` `release_version` `content_hash`
- lesson/page/section anchors
-
- rubric
-
- `/provider` embedding signature

“ ”

fail closed

7.3

lesson / chunk “ ”“ ”“ ”
answerability gate

7.4 RAG

Recall@K MRR/nDCG / —
p50/p95 / baseline
authoring corpus Tutor development/eval cases
judge rubric Tutor Tutor TutorBench judge Tutor
ground truth [P2]

8.

8.1

Living Book Co-Writer
DeepTutor anchor

typed block
fail open

“ — — — — ”
[D9]

8.2

retrieve read_source grade run_controlled_check save_observation
Practice Authoring

Explain Coach
run_controlled_check

8.3

Tutor “ / / ” DeepTutor /

token

8.4

SYSTEM/OS

[D10]

Memory

8.5

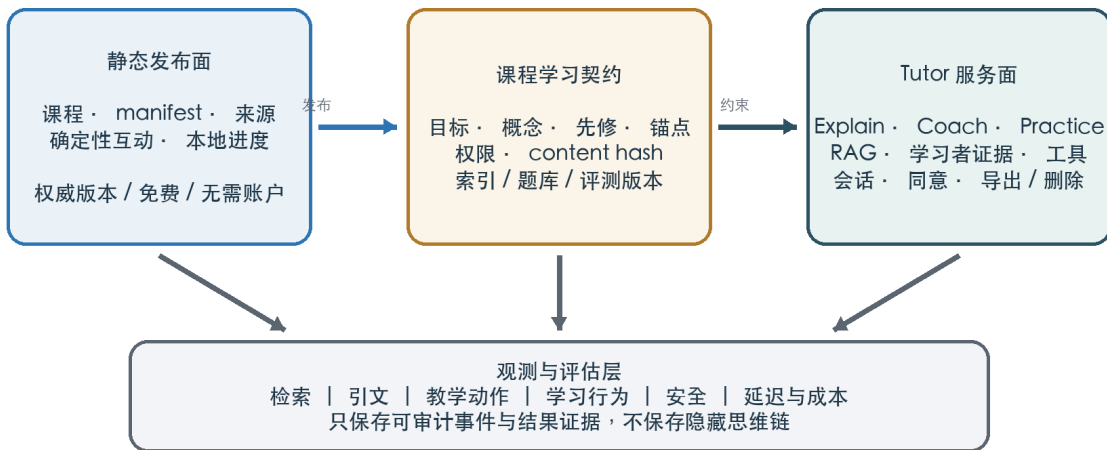
DeepTutor Skills/MCP/CLI
schema

“ ”“ ”“ ”

9.

推荐目标架构：双平面 + 课程学习契约

静态课程保持权威与低运维；Tutor 作为独立、可选、可撤销的服务运行。



1 Tutor

9.1

manifest	Next.js	Tutor	Tutor
[L2] [L3]			

9.2 Tutor

profile / RAG / anonymous_session_id / RAG

9.3

hash	Tutor	course_version	contract_hash
fail closed	Tutor	feature flag	kill switch

9.4

/ Tutor / authoring Tutor

10.

2 0.5 / /

□□	□□□□□□□	□□□□	□□□□□
P0-0	RAG 50-100 20-30 Tutor cases Tutor	/hash fail closed	RAG 3-4
P0-1 MVP	observation Memory	feature flag/kill switch /	/ 6-8
P1-2a		/	/ 4-6
P1-2b	Evidence/Correctness/Pedagogy gates outcome set	/	10 6-
P1-3	proposal/spine/typed blocks diff S00-S10	AI QA	Tutor 6-1 0
P2-4	PageIndex/	A/B /	

DeepTutor A

DeepTutor /

11.

11.1

/
“RAG”
Recall@K/MRR/nDCG
/
p95
baseline

11.2

rubric
LLM judge
DeepTutor
human alignment
[P2]

11.3

“”“”“
Tutor”
/
“”

12.

□□	□□□□□	□□□□	□□□□/□□
	DeepTutor	RAG SLO	/ Tutor
		/	
	QuestionPipeline schema r epair	Evidence/Correctness/Pedagogy gates	validator
	provider	A/B	
	Tutor /		
	retention/ /		[P2]
	Apache-2.0	LICENSE/NOTICE SBOM / / / /	[D11]

DeepTutor
A

OQ Naive Tutor 10.76% “ 10.76%” [P2]

13.

	RAG	50-100	20-30	Tutor
cases	Memory		/ / /	
	DeepTutor			

A

DeepTutor

- [D1] HKUDS/DeepTutor v1.5.16 README Book KB Memory [README L203–213](#)
- [D2] [v1.5.16](#)
- [D3] Tools/Capabilities [AGENTS.md L1–71](#)
- [D4] provider KB [RAG factory L1–50](#)
- [D5] BM25 RRF [LlamaIndex retriever L119–151](#)
- [D6] embedding signature `needs_reindex` [LlamaIndex pipeline L126–226](#)
- [D7] Memory L3 [trace.py L1–67](#) [L3 prompt L1–37](#)
- [D8] Explore/Plan/Quiz [QuestionPipeline overview](#) [schema issue checks](#)
- [D9] proposal spine typed blocks anchors [Book models L176–278](#)
- [D10] SYSTEM/APPLICATION/OFF [Sandbox spec L16–40](#)
- [D11] [Apache-2.0 LICENSE](#)
- [D12] [Mastery scoring L1–37](#)

DeepTutor

- [P1] *DeepTutor: Towards Agentic Personalized Tutoring* v3 2026-07-09 [arXiv](#)
- [P2] TutorBench [arXiv v3 PDF](#)
- [L1] `README.md:1–7, 11–30, 72–88, 156–180` README / RAG /
- [L2] `next.config.ts:3–23` `output: "export"`
- [L3] `lib/progress.ts:1–52` `localStorage-only`
- [L4] `lib/deepseek.ts:1–11, 76–122` `components/lab/Lab.tsx:32–49, 85–142` BYOK
- [L5] `lib/rag/manifest.ts:9–278` `lib/rag/types.ts:48–126, 273–320` RAG 4 12 34 concepts 40 sources
- [L6] `components/rag/RetrievalLab.tsx:8–79` RAG

- [L7] lib/courses.ts:290–450, 510–554 available Cursor soon
- [L8] course_review_2026-08-23/staging/AI_V3.1_/manifest.json S00–S10
- [L9] messages/cursor/en.json:581–618 Cursor Lesson 13

B

- DeepTutor GitHub
- v3 PDF LLM judge
- 2026-08-23 NOT_ASSESSABLE available
- baseline /